

Analysis on the Real Line HW 2

Robert O'Connell

4th February 2026

BS - Page 30 Exercise 18

Let $a, b \in \mathbb{R}$, and suppose that for every $\epsilon > 0$ we have $a \leq b + \epsilon$. We want to prove that $a \leq b$.

To do this we will use a contrapositive proof.

i.e. prove there exists an ϵ such that $a > b \Rightarrow a > b + \epsilon$

Since $a > b \Rightarrow a - b > 0$.

Picking epsilon to be $\frac{a-b}{2}$, which is positive, we see that $a = b + 2\epsilon$

Thus $a > b + \epsilon$

Which proves that $a \leq b + \epsilon \Rightarrow a \leq b$

BS - Page 34 Exercise 5

We know $a < x < b$ and $a < y < b \Rightarrow -b < -y < -a$.

Adding these together we get $a - b < x - y < b - a$

Since $-(a - b) = b - a$, we can use one of the properties of the absolute value function to get $|x - y| < b - a$.

Interpreting this geometrically, we get that the distance between a and b is always greater than the distance between x and y

BS - Page 38 Exercise 5

Define $-S = \{-s | s \in S\}$

Let s_0 be $\inf(S)$

- $s_0 \geq s \quad \forall s \in S$
- If z is another lower bound for S , i.e. $z \geq s \quad \forall s \in S$, then $s_0 \geq z \quad \forall z$

Using these properties we see

- $-s_0 \leq -s \quad \forall s \in S$
- $-s_0 \leq z \quad \forall z$ such that $-z \leq -s \quad \forall s \in S$

Then

- $-s_0 \leq k \quad \forall k \in -S$
- $-s_0 \leq z \quad \forall z$ such that $-z \leq k \quad \forall k \in -S$

Thus we see $-s_0$ satisfies the properties of being a supremum on the set $-S$

$$-s_0 = \sup(-S) \Rightarrow \inf(S) = -\sup(-S)$$

BS - Page 43 Exercise 1

From class we have that $\sup(x + A) = x + \sup(A)$, $A \subset \mathbb{R}, x \in \mathbb{R}$
i.e. $\sup(1 - \frac{1}{n} | n \in \mathbb{N}) = 1 + \sup(-\frac{1}{n} | n \in \mathbb{N})$

Now we need to show, $\sup(-\frac{1}{n} | n \in \mathbb{N}) = 0$

- $n > 0 \Rightarrow -\frac{1}{n} < 0 \quad \forall n \in \mathbb{N}$
0 is an upper bound for the set

- Let $s = \sup(-\frac{1}{n} | n \in \mathbb{N})$

Hence $s \leq 0$

If $s < 0 \Rightarrow -s > 0$ then by the Archimedean Property

- $-sn > 1$ for some $n \in \mathbb{N}$
- $-s > \frac{1}{n}$
- $s < -\frac{1}{n}$
- s is not an upper bound, a contradiction

Hence $s = 0$

$$\sup(1 - \frac{1}{n} | n \in \mathbb{N}) = 1 + 0 = 1$$